

Customer Churn — Exploratory Data Analysis

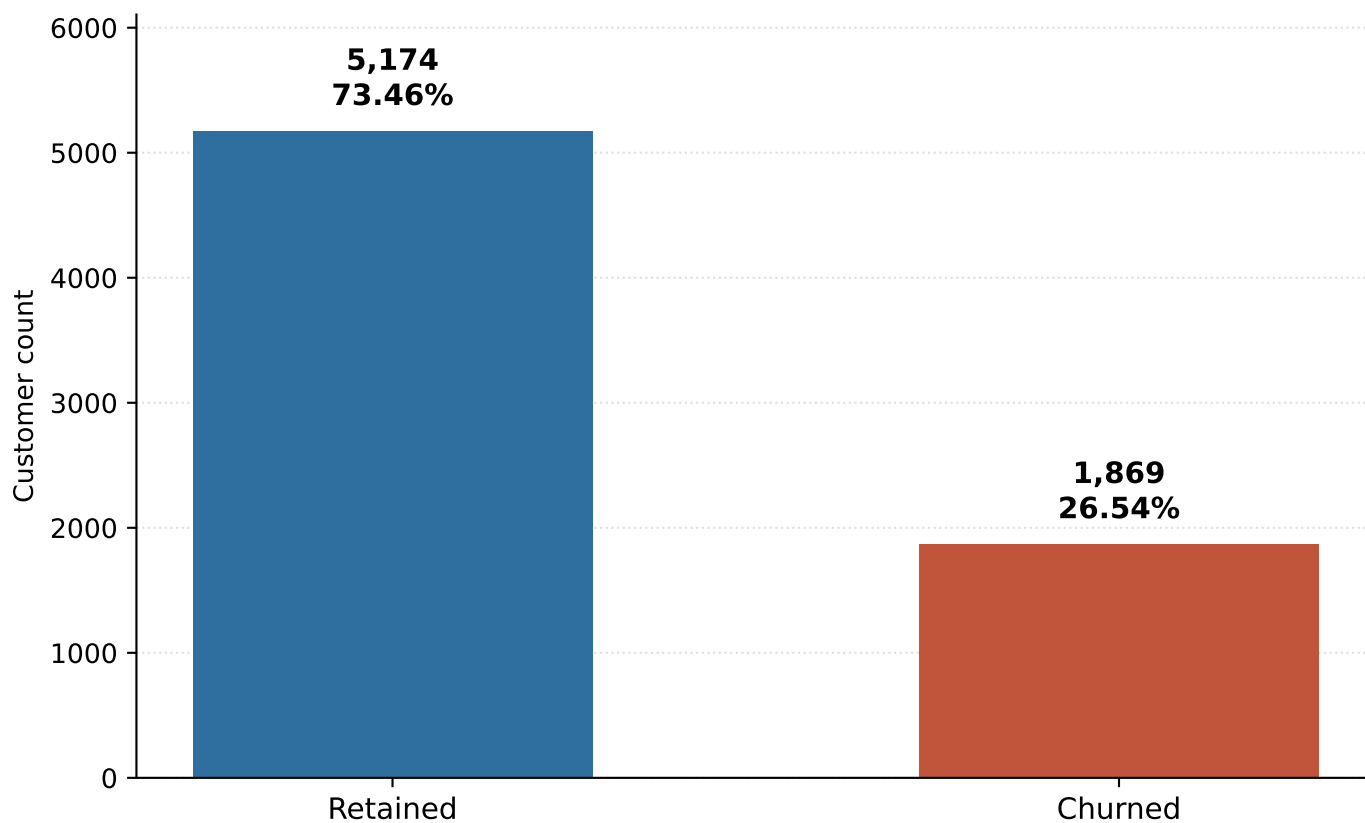
1. Dataset Overview

Dataset:	data/raw/WA_Fn-UseC_-Telco-Customer-Churn.csv
Total records:	7,043
Target variable:	Churn
Analysis scope:	Investigation 1 – retention vs. churn baseline Distribution of the target variable. Investigation 2 – tenure vs. churn Tenure distributions of the two churn populations, described only. No other variable is examined in this report.
Data treatment:	Read-only. The raw dataset is not modified, cleaned, or transformed by this analysis.

2. Retention vs. Churn

Counts and percentages are computed from the loaded dataset.

Customers by status



Customer Status	Customer Count	Percentage
Retained	5,174	73.46%
Churned	1,869	26.54%
Total	7,043	100.00%

3. Interpretation — Retention vs. Churn

Of the 7,043 customers in the dataset, 5,174 customers (73.46%) are recorded as retained (Churn = "No") and 1,869 customers (26.54%) are recorded as churned (Churn = "Yes").

Retained customers therefore outnumber churned customers by 3,305 customers, a gap of 46.93% of the dataset.

Every record carries one of the two documented target values, "No" or "Yes"; the target contains no missing values and no other category. The two percentages above therefore account for the whole dataset.

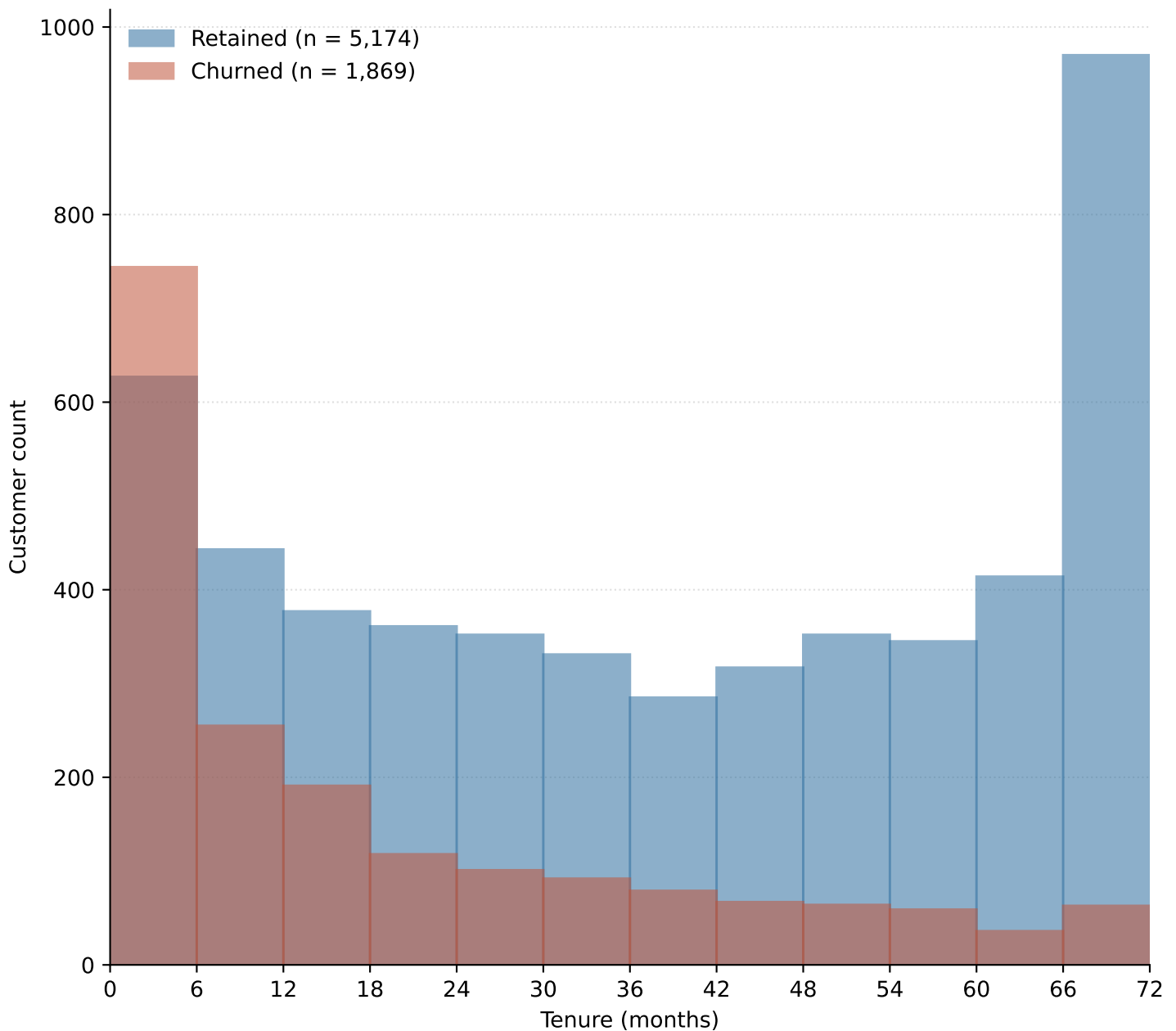
These proportions are the baseline for the exploratory analysis that follows. Subsequent investigations compare the churn rate within individual segments of the data against the overall rate reported here, and the same proportions describe the class balance any later model would be trained against.

This investigation examines the distribution of the target variable only. It establishes how many customers churned, not why: no relationship between churn and any other variable has been measured at this stage, and no cause is inferred from these counts.

4. Tenure Distribution by Churn Status

Every recorded tenure observation is plotted. The 6-month bars display the continuous distribution; they are not analytical tenure bands, and no churn rate is computed for them.

Customers by tenure, in 6-month display bins



5. Tenure Distribution by Churn Status — Box Plot

Drawn from the same tenure observations as Section 4. Boxes span the interquartile range, the line inside each box is the median, and points beyond the whiskers are shown rather than removed.

